

ITD – README

1. Project Title

ITD X DDA ASG1 – PetPals AR

2. Description of the Game / Application

This Unity project is a first-person interactive experience featuring AR Virtual Pets. Players scan AR pet posters, interact with a pet system (petting and feeding), and do actions to evolve their pet.

Main features include:

- AR poster scanning to spawn pets
 - Pet system (happiness/XP level, hunger, levelling up)
 - Toys interaction system
 - Pet death system
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3. Installation & Setup Instructions

3.1 Requirements

Component	Requirement
Platform	Android Phone / iPhone
Hardware	A smartphone capable of running AR applications
Controls	Touchscreen
Engine	Unity 2022/2023 LTS

3.2 How to Run the Game

1. Download the *APK* (Android) or *iOS TestFlight build* .
 2. Install and launch the app.
 3. Allow camera permissions.
 4. Point your camera at a pet AR poster to spawn your pet.
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4. Controls

Mobile

Action	Control
Look Around	Move phone camera around

Action	Control
Scanning pet	Scan a pet poster
Interact with pet	Tap the Interact icon on pet
Pet the Pet	Tap the "Pet" button/icon in the interact menu
Feed the Pet	Tap the "Feed" button/icon in the interact menu
Use a Toy	Open 3 line menu, press "Toys Owned", Tap "Use Now" → drag toy around with your finger
Close Tutorial	Tap the <i>X</i> button on any tutorial page
Next Page (Tutorial)	Tap <i>Next</i>
Open Sidebar Menu	Tap the <i>3 line menu icon</i> ($\equiv$)

Mouse and Keyboard

Action	Key
Move	<i>W A S D</i> + Hold right mouse
Interact with pet	Click on Interact Icon
Feed pet	Click "Feed" Icon
Pet the pet	Click "Pet" Icon
Use Toy	Drag with mouse
Close Tutorial	Click <i>X</i>

5. Gameplay Walkthrough

5.1 Tutorial System

- The tutorial appears on the first launch.
- Press *Next* to move through the pages.
- Press *Start Playing* to begin the game.
- Click *X* on any page to skip the tutorial.
- You can reopen it anytime via *3-line Menu* → *Tutorial*.

5.2 Objective: Evolve at least 1 pet

Your goal is to:

1. Increase the pet's happiness level
2. Reach pet happiness level 10 to unlock the evolution
3. Evolve the pet

6. Pet System

6.1 Happiness

- Petting increases happiness by +1.
- Toys increase happiness based on toy type.
- When happiness reaches the required amount, the pet levels up.

6.2 Hunger

- Every pet has a hunger bar
- Hunger bar decreases by *1 bar per minute*.
- Feeding restores *1 bar*.
- If hunger reaches 0, the pet dies.

6.3 Death

- If hunger bar reaches 0, a 3 second timer begins.
 - If the pet is not fed within these 3 seconds the pet dies and is permanently gone.
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7. Toy System

How Toys Work

1. Open the *Toys Owned* page.
 2. Press *Use Now* on any toy to spawn the toy model.
 3. Drag the toy around the scene.
 4. When the toy touches the pet, the toy disappears.
 5. The pet gains happiness.
 6. The toy enters a cooldown of 45 seconds before it can be used again.
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8. Known Limitations & Bugs

- Game progress does not save when exiting and logging in back to the game.
 - Petting and feeding particles do not appear when petting and feeding the dragon.
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9. Credits & Assets Used

3D Models

- Pet models (cat, dog, dragon) – SketchFab
- Toy models (Sackboy, bone, ball of yarn) - SketchFab